

On Causative Recursion: A Response to Key 2013 and Nie 2025

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1. Introduction

Key (2013) argues that Turkish causatives are not recursive despite allowing more than one causative morpheme on a single verb. Empirically, there is not always one-to-one corresponding between Turkish causative morphology and interpretation (Key 2013: 224; Nie 2025 402–403). Both “vacuous causatives”, in which the causative morphology lacks corresponding causative meaning, and “extended causatives”, in which the causative meaning lacks corresponding causative morphology, are found in Turkish, according to Key (2013). Nie (2025) confirms this.

(1) Saç-im-i kes-tir-t-ti-m. (vacuous causative)

hair-1SG-ACC cut-CS-CS-PST-1SG

‘I had my hair cut.’ (Key 2013: 221)

(2) Anne-si Ayşe-ye Mary-yi oyna-t-ti. (extended causative)

mother-3SG.POSS Ayşe-DAT Mary-ACC play-CS-PST

‘Her mother made Ayşe make Mary play.’ (Nie 2025: 403)

Based on these, Key (2013) claims that Turkish causatives along with those in other languages such as Hungarian, Kashmiri, and Tsez encode at most two events, one causing and another caused, in the syntax and that the morphological redundancy for vacuous causatives and the morphological lack for extended causatives remain superficial phenomena, which creates the illusion of syntactic recursion. Theoretically, Key (2013) argues that there is a dedicated Caus head for the linking between so-called causing event and the causee, with the causer introduced by Voice postulated above Caus. That is, there is no recursion of a given syntactic category, for Key (2013). Key’s (2013) analysis thus follows the classic “cause-become” spirit, where relevant semantics is encoded by syntactic heads in the lexicon.

Nie (2025), however, argues against Key (2013) by providing both empirical and theoretical reasonings. She observes that Turkish causatives can be syntactically recursive as tested out by her scope-of-again test, manner-adverb test, temporal-adverb test, and permission-reading test. While all the tests seem valid, the most important one is the manner-adverb test

because positions and scope of manner adverbs are more severe than other types of adverbs. According to this test, it is possible for the manner adverb *sessizce* ‘quietly’, variable in position, to take a scope over any of the two “causing events” and the caused event.

(3) Anne-si Ayşe-ye *sessizce* Mary-yi oyna-**t-tir**-di.
 mother-3SG.POSS Ayşe-DAT *quietly* Mary-ACC play-CS-CS-PST
 ‘Her mother made Ayşe make Mary play quietly.’

- a. *quietly* CAUS > CAUS > V
- b. CAUS > *quietly* CAUS > V
- c. #CAUS > CAUS > *quietly* V

(4) Anne-si Ayşe-ye Mary-yi *sessizce* oyna-**t-tir**-di.
 mother-3SG.POSS Ayşe-DAT Mary-ACC *quietly* play-CS-CS-PST
 ‘Her mother made Ayşe make Mary play quietly.’

- a. *quietly* CAUS > CAUS > V
- b. #CAUS > *quietly* CAUS > V
- c. CAUS > CAUS > *quietly* V

This leads Nie (2025) to claim that recursive events are denoted by Turkish causatives and that this is correlated with syntactic causative recursion. As for how to account for this, Nie (2020, 2025) argues that Turkish causatives have a structure where a Voice head is stacked by another, without appealing to dedicated heads such as Caus and *v*.

According to the empirical results in Key (2013) and Nie (2025), Turkish causatives are recursive, which is solidly evidenced by the manner-adverb test by Nie (2025). This points to Nie’s (2025) Voice-over-Voice analysis, but not Key’s (2013) “Caus-over-*v*” analysis.

What remain unclear from the two studies are the following. First, Turkish, along with Hungarian, Kashmiri, and Tsez as well as Japanese, reported in Key (2013: 225), allow morphology-syntax mismatch. Key (2013: 225) attributes the mismatch to a morpho-phonological factor, arguing that causatives in these languages are not syntactically recursive. While Nie (2025) correctly observes that causatives are indeed syntactically recursive, her analysis fails to show that why the mismatch is allowed. Second, a seemingly trivial but actually vital problem as to why modification by manner adverbs is more restricted than that by other types of adverbs, briefly mentioned by Nie (2025: 407). Note that in (3), the adverb cannot modify the “caused” event and in (4), it cannot modify the lower “causing” event.

This squib is devoted to solving these two problems based on relevant facts about Mongolian causatives in comparison to Turkish causatives. We first overview the basics of

Mongolian causatives in Section 2 and discuss how the problems can be solved.

2. Khalkha Mongolian causatives

In order to keep our discussion as compact as required for a squib, we only overview relevant and important facts about the Khalkha dialect of Mongolian (henceforth, Mongolian). Mongolian, another morphological language, has three categories of causative morphology. They are *AA*, *LGA* and *UUL*, including their allophonic and/or allomorphic variants, which comply with vowel harmony.¹

Among the three categories, *AA* never follows the other two and *LGA* never follows *UUL*, for phonological and syntactic reasons.² What concerns us most in this squib is that *LGA* and *UUL*, attached to V, create either bi-eventive or tri-eventive causatives. Normally, transitive-based *LGA-UUL* causatives are tri-eventive, in which four arguments are present, as exemplified below.

- (5) ... Dorž-oor hüü-d-ee em-($\emptyset$)-ni uu-lg(a)-uul-la.
Dorž-INS son-DAT-RX medicine-ACC-PSS[3] drink-CS-CS-PST
‘With Dorž’s help, I made my son drink the medicine.’

(Kullman and Tserenpil 2015: 119)

Thus, for *LGA-UUL* causatives to denote bi-eventive causation, exemplified in (6b), is expected to be surprising given that bi-eventive causatives normally allow a single causative morpheme. That is, the *LGA-UUL* causative in (6a) and the *LGA* causative in (6b) are essentially identical regarding their syntax and semantics despite the number of causative morpheme varies.

- (6) a. Bi egč-eer nig beeli hii-lge-sen.
b. Bi egč-eer nig beeli hii-lge-uul-sen. (ibid.: 119)³
‘1SG-NOM lady-INS one glove make-CS(-CS)-PST’
‘I had a lady make me a pair of gloves.’
*‘I had a lady make someone make me a pair of nice gloves.’

Causatives such as (6) would be typical examples of “vacuous causative” in Key’s (2013)

¹ See Janhunen (2012: 249–250) and Kullmann and Tserenpil (2015: 117–123), among many others, for more basics of Mongolian causative morphemes.

² See Bai (2025) for detailed discussion on this.

³ Note that the applicative *me* in the translation is not an element corresponding to a causee in any case. It instead reflects the implicit presence of a reflexive nominal referring to the subject, as indicated by the fact that the reflexive pronoun *öör* ‘self’ can be added to the sentence. Note that the dative causee in normal *LGA-UUL* causatives as in (5) is not interpreted in this way; it is not reflexive. See Kullmann and Tserenpil (2015: 119) for more examples of the same type.

terms.⁴ However, the suffix *UUL* in these causatives is not vacuous *per se*. The reason for this is as follows. Unlike the tri-eventive *LGA-UUL* causative in (5), which contains a dative causee, the abnormal *LGA-UUL* causative in (6b) lacks a dative causee. Notably, there is neither an implicit causee nor an omitted one in (6b);⁵ that is, the seemingly vacuous causatives involve only three syntactically present arguments, lacking a dative causee, which is otherwise required in normal *LGA-UUL* causative as in (5). Instead, the abnormal *LGA-UUL* causative in (6b) is interpreted as the *LGA* causative in (6a), where a single suffix is used. This means that *UUL* in (6b) has no associated nominal argument, which makes it look like a vacuous morpheme. Thus, abnormal, bi-eventive *LGA-UUL* causatives are not syntactically recursive despite the recursion of causative morphology, unlike normal, tri-eventive *LGA-UUL* causatives, which are recursive both syntactically and morphologically, complying with the Mirror Principle (Baker 1985).

So far, Mongolian has vacuous causatives on the surface, just like Turkish and other languages mentioned in Section 1. However, this is not the case, as we will see in Section 3.

In what follows, we overview facts about modification by manner adverbs in recursive causatives. As in (7–9), a manner adverb can appear in three different positions but their modification scopally varies among the positions. When immediately following the highest causer, *uurtaigaar* ‘angrily’ can take a scope only over the event associated with the causer, as in (7); when following the highest causee, it can attach at two points at LF, as sketched in (8); and when following the causee, it can attach at two points, too, as sketched in (9). But note that in the latter two cases, the second attaching point is reversed. This is essentially the same pattern as that Nie (2025: 406) observes for Turkish causatives as in (3–4).

(7) Zahiral *uurtaigaar* *bagš-aar* *surugč-ig* *ir-үүл-үүл-sen*.
principal-NOM *angrily* teacher-INS student-ACC come-CS-CS-PST
‘The principal made the teacher make the student come angrily.’

- a. angrily CAUS > CAUS > V
- b. *CAUS > angrily CAUS > V
- c. *CAUS > CAUS > angrily V

(8) Zahiral *bagš-aar* *uurtaigaar* *surugč-ig* *ir-үүл-үүл-sen*.
principal-NOM teacher-INS *angrily* student-ACC come-CS-CS-PST
‘The principal made the teacher make the student come angrily.’

⁴ See Kullmann and Tserenpil (2015: 119) for more examples of vacuous causatives.

⁵ An implicit causee is available in other types of causatives such as *uz-uul* ‘show; literally, cause to see’ and *zas-uul* ‘have ... repaired; literally, cause to repair’.

- a. %angrily CAUS > CAUS > V
 - b. CAUS > angrily CAUS > V
 - c. *angrily CAUS > CAUS > V
- (9) Zahirāl bagš-aar surugč-ig **uurtaigaar** ir-үүл-үүл-sen.
principal-NOM teacher-INS student-ACC **angrily** come-CS-CS-PST
‘The principal made the teacher make the student come angrily.’
- a. angrily CAUS > CAUS > V
 - b. *CAUS > angrily CAUS > V
 - c. CAUS > CAUS > angrily V

This strengthens Nie’s (2025) analysis of causative recursion on the one hand and sheds light on the nature of manner adverb modification interacting with causative recursion. We return to this issue in Section 3.

3. Discussion

We tackle, from a theoretical perspective, two primary problems in this section, one concerning the morphology-syntax mismatch and another concerning the manner adverb modification. The first problem is tackled based on observations regarding morpho-phonological assimilation of Mongolian causative suffixes *LGA* and *UUL*, which leads to the non-recursive property of *LGA-UUL* causatives as in (6b).

To begin with, the actual pronunciation of the *LGA-UUL* string does not involve the end vowel *A* of *LGA*, rendering the string monosyllabic: *LGUUL*. This is more straightforward given that *UUL* has a variant *GUUL*, which is used for verbs ending with a long vowel such as *gii-güül* ‘illuminate; literally ‘cause to dawn’ and those with a diphthong such as *bai-guul* ‘build; establish; literally, cause to be’.

A note about the orthographies of Mongolian is necessary to clarify this. While *LGUUL* is written as “*лгуул*” in the Cyrillic orthography, which is a single syllable reflecting the assimilation, it is written as “*лгуул*” in the Uyghur orthography, which is disyllabic. Notably, in colloquial Mongolian, the disyllabic “*лгуул*” (*LGA-GUUL*) is normally not monosyllabically, not disyllabically, pronounced; thus, it is pronounced in the same way as the Cyrillic “*лгуул*” (*LGUUL*). However, in some eastern colloquial Inner Mongolian dialects, “*лгуул*” (*LGA-GUUL*) is still disyllabically pronounced, reflecting the disyllabic structure serving as the base for the assimilation. All this points to the fact that *GUUL* in *LGUUL* is not an accident and that *LGUUL* is in fact a newly emerging suffix, which is led by the fact that the Khalkha dialect underwent a script reform in the 1940s.

As such, the disyllabic *LGA-UUL* is the full-fledged form of the monosyllabic *LGUUL*, both functioning to denote a single causation. This means that *LGA-UUL* causatives, now *LGUUL* causatives, are in the stricter sense not vacuous and hence non-recursive and that Mongolian do not have true vacuous causatives.

This opens the possibility that where such assimilation, either phonological or morphological, between causative suffixes is available, the redundancy of causative morphology may not necessarily lead to morphology-syntax mismatch. In what follows, we look at if this is the case for Turkish causatives.

The double causative suffixes used in the Turkish causative sentences in Key (2013) and Nie (2025) include *t-tur*, *t-tir*, *tur-t*, and *tir-t*. Admittedly, on the surface these do not exhibit the assimilation-related properties like those observed for the Mongolian *LGA-UUL* as *GUUL*. However, it is quite straightforward that all these Turkish causative suffixes contain a consonant *t* in each of a given pair of causative suffixes. Interestingly, Hungarian *at-tat* and Tsez *r-er* as well as Japanese *sase*, which Key (2013: 224–225) believes to be counterevidence for syntactic recursion of causatives cross-linguistically, exhibit similar properties to Turkish suffixes mentioned above. This is presumably not coincidental. Key (2013) takes these as examples of morphology-syntax mismatch to support his “Caus-over-v” analysis.

Although any assimilation related to these double suffixes is not easy to properly formulable for the time being, it seems reasonable to not take them as evidence for the morphology-syntax mismatch under discussion and a non-recursive analysis of morphological causatives, at least those in Turkish.

Nie (2025) reports from Kural (1996: 126) and Göksel (1993) that there is no upper limit as to the number of the iterated Turkish causative morpheme, providing *doğra-t-tir-t-tir-t* ‘cause to chop’ as an example. I presume that this also has to do with the phonological properties of the causative suffixes *t* and *tir*, due to which artificial iteration of them becomes possible. It is thus reasonable that the vacuous use of causative morphemes is possible when they exhibit certain phonological properties including distributions of a consonant in two separate suffixes and probably affixal compactness as well, and that so-called “extension causatives” are possible only when the causative suffix itself contains two occurrences of a single consonant as in Japanese *sase*.⁶ As such, so-called “vacuous causatives” and “extension causatives” are not arbitrary, although they are primarily a phonological phenomenon. Importantly, however, they

⁶ See Key (2013: 225) and Miyagawa (1999).

cannot be used as evidence for syntactic non-recursion of causatives. Syntactic recursion of causatives is not an illusion but rather morphology reduplication/doubling of causatives is.

Now we turn to what manner adverb modification tells us about the nature of morphological causatives. As can be seen from our observation about Mongolian in Section 2 and Nie’s (2025) observation about Turkish, the scope-related semantics of manner adverb modification is constrained by three interventions: Hierarchical intervention (HI) — predicate-oriented intervention, nesting intervention (NI)⁷ — subject-oriented intervention, and linear intervention (LI) — subject-oriented intervention. Importantly, HI, being hierarchical, dominates the other two, being non-hierarchical. Specifically, as represented below, for a manner adverb to take a scope over a given projection, HI must fail, that is, adverb modification must not be blocked by HI. Where this is guaranteed, either NI or LI must also fail. If neither NI nor LI fails, the adverb cannot take a scope over a relevant projection despite the failure of HI. Let us elaborate on this. Use the bar-level representation to indicate the projection or node to be under the adverbial scope at LF, with irrelevant nodes unexpressed in (10–12).

We first look at the case of the adverb immediately following the causer, notated as *sbj3*. Modification of the event-core (*V'*) is impossible because HI by Caus-1, for the first, lower causation, as well as HI by Caus-2, for the second, higher causation, does not fail. Modification of Caus-1 (*Y'*) is also impossible because HI by Caus-2 (*X'*) does not fail. Modification of Caus-2, in contrast, is possible because HI, NI, and LI all fail.

(10) Syntax: ... *sbj3* **adv** [*x'* *sbj2* [*y'* *sbj1* [*v'* ...
 LF: **adv**(√) Caus-2 **adv**(×) Caus-1 **adv**(×) Event-core

Now we look at the case of the adverb following the causee in the second round of causation, notated as *sbj2*. Modification of *V'* is impossible because HI by Caus-1 does not fail. Modification of Caus-1 is possible because HI and LI (one of the two subject-oriented interventions) fail. Modification of Caus-2 is possible, too, because HI and NI (the other of the two subject-oriented interventions) fails.

(11) Syntax: ... *sbj3* [*x'* *sbj2*] **adv** [*y'* *sbj1* [*v'* ...
 LF: **adv**(√) Caus-2 **adv**(√) Caus-1 **adv**(×) Event-core

In the third case, the adverb linearly follows the causee in the first round of causation, notated as *sbj1*, which is the subject of the base verb. Modification of *V'* is possible because HI and LI (one of the two subject-oriented interventions) fail. Modification of Caus-1 is impossible

⁷ See Section 4 for detailed discussion.

because none of HI, LI, and NI fails. Modification of Caus-2 is possible because HI and NI (the other of the two subject-oriented interventions) fail.

(12) Syntax: $\dots \text{sbj3} \quad [X' \quad \boxed{\text{sbj2} \quad [Y' \quad \boxed{\text{sbj1}]}] \quad \text{adv} \quad [V' \quad \dots]$
 LF: $\text{adv}(\surd) \text{Caus-2} \quad \text{adv}(\times) \text{Caus-1} \quad \text{adv}(\surd) \text{Event-core}$

The interaction between HI and one of NI and LI regarding adverb modification shows that the one-to-one correspondence between the hierarchical projections and the arguments, which I view as potential subjects,⁸ interact with each other for allowing and/or disallowing modification of the projections by manner adverbs.

From all this, it follows that stacked functional heads in morphological causatives are not the locus of so-called causing events, nor do they introduce them; they themselves have no semantic content such as eventivity. They are instead simply sbj-introducers, linking a sbj and its sister — a bar-level predicate. In this sense, causative morphemes do not denote an event but denote a subject-predicate relation, which we call “causation”. What is modified by a manner adverb is thus an event core (V’ or V) and/or the constellation “relation + event (core)”, which creates a larger event. Larger events are represented by X’ and Y’ in (10–12), which are bar-level functional projections, namely, two instances Voice’, hence Voice-over-Voice (Nie 2020) or Split Voice (Bai 2025).

Thus, strictly speaking, there does not exist so-called “causing event” and hence nor does so-called “caused event”, at least in Mongolian and Turkish type morphological causatives,⁹ while we are used to interpreting the causation relation as a type of abstract event since Shibatani (1976). I, however, have used “causing event” and “caused event” as terms in this squib, just for expository purposes, without theoretical implications.

This is why modification by manner adverbs is constrained by HI, LI, and NI, that is, why manner adverbs cannot modify intervened projections. Specifically, given that manner adverbs take a scope over a projection, not over a head or a specifier, intervention (HI) plays a role where necessary. HI does not block modification of a head or a specifier, if any. If a functional head has eventive semantics and if a causative morpheme denotes a so-called causing event on its own right, it would be the case that manner adverb modification is irrelevant to HI and associated ones such as NI and LI; that is, manner adverbs would freely modify a verbal head

⁸ The causer and causee are in fact potential (predicate-internal) subjects, as observed by Bai (2025).

⁹ Note that in English periphrastic causatives, a causative verb such as *make* has a semi-lexical meaning, indicating that it encodes eventive semantics in some form.

from more variable positions, being less restricted.¹⁰ But this is not the case.

4. Conclusion

An important consequence of the discussion in Section 3 is that morphological causatives are indeed derived over a Voice-stacking structure, which involves multiple occurrences of a single type of syntactic head, namely, Voice (Nie 2020; Sigurðsson and Wood 2021; Bai 2025). It is not the case that recursive causatives involve hierarchically fixed functional heads; causative recursion cannot be attributed to a dedicated Caus head plus other types of heads as often assumed by studies such as Pylkkänen (2008), Harley (2013), Key (2013), and many others.

Morphological causatives, at least in Mongolian and Turkish, denote “overlapping” rather than “superposing” events. By “overlapping”, I mean that an event contains another, creating an event-nesting, and by “superposing”, I mean that an event leads to another, creating an event-array. For the former, causation is simply a relation between an event and a nominal, whereas for the latter, it is a relation between two events, where an event takes place before another, as stipulated by Shibatani (1976) and many.¹¹ When the overlapping event structure is extended by addition of nominals, named causers or subjects, it, mapped into the syntax, is identified in the form of stacked structure of a functional category, which is built in a hierarchical manner. Specifically, VP represents the “inner” event, VoiceP1 represents the “outer” event, and VoiceP2, when present, represents an outermost event.¹² An outer(most) event takes place by virtue of the arrival of a nominal, named “causer”, thereby having an inner event nested inside it. The nesting structure thus plays a role in the interactions among the syntax, semantics, and morphology of causatives. The manner adverb modification we discussed in Section 3 is a good

¹⁰ It is noted that in Turkish causatives, temporal adverbs and *again*-adverbs attach more points than manner adverbs at LF, as observed by Nie (2025: 405–408).

¹¹ Shibatani’s (1976) interpretation of causatives can be paraphrased by, for example, “John did something, which caused the door to become open”, where “doing something” is named “a causing event” and “the door’s opening” is named “a caused event”. This interpretation is applied to, for example, “John opened the door”, where the causing event is assumed to be implicitly present. However, if this was the case, it would remain unclear why it is never linguistically realized as lexical verbs (not functional affixes) in morphological causatives. It has been believed that adverbs are the sign of eventivity and adverbial modification has often been used as diagnostics for it. However, it is not necessarily the case that adverbial modification is linked with eventivity. Note that adverbs are often used to modify non-verbal and non-eventive phrases. On the other hand, causation, which is not an event but rather a linkage between a nominal and an event (“caused event”, conventionally speaking), is also subject to adverbial modification. It is noted that stative verbs, which normally do not denote events, are readily modified by adverbs, which often modify eventive verbs, e.g., *People intentionally and deliberately are distant or pushy to make a point*, cited from COCA.

¹² This said, a head such as Voice or any dedicated one such as Caus does not represent so-called “causing” event in the strict sense, contrary to the “cause-become”-based approach to morphological causatives.

example of this. Importantly, the nesting structure, being mapped to VoiceP-stacking in the syntax, also leads to syntactic and semantic variations related with subjects, transitivity, and grammatical voice, an issue to address in a separate paper. Consequently, the derivation of recursive causatives readily comes down to the merger of causers and VoiceP, which is allowed and demanded by Free Merge and simplism elucidated by Chomsky (2013, 2015, 2024), assuming as few as possible ingredients.

References

- Bai, Chigchi. 2025. Mirror Principle and Split Voice: Evidence from Mongolian and Japanese. *Asian Languages and Linguistics* 6:253–286.
- Baker, Marc. 1985. The mirror principle and morphosyntactic explanation. *Linguistic Inquiry* 16:373–415.
- Chomsky, Noam. 2013. Problems of Projection. *Lingua* 130:33–49.
- Chomsky, Noam. 2015. Problems of projection: Extensions. In *Structures, Strategies and Beyond: Studies in Honor of Adriana Belletti* (eds.), Elisa Di Domenico, Cornelia Hamann and Simona Matteini, 3–16. Amsterdam: John Benjamins.
- Chomsky, Noam. 2024. The Miracle Creed and SMT, In G. Bocci, D. Botteri, C. Manetti and V. Moscati (eds.), *Rich Descriptions and Simple Explanations in Morphosyntax and Language Acquisition*, 17–39. Oxford: Oxford University Press.
- Davies, Mark. 2008-. The Corpus of Contemporary American English (COCA): 520 million words, 1990-present. <https://www.english-corpora.org/coca/>.
- Göksel, Aslı. 1993. Levels of representation and argument structure in Turkish. Doctoral dissertation, University of London.
- Harley, Heidi. 2013. External arguments and the Mirror Principle: On the distinctness of Voice and v. *Lingua* 125:34–57.
- Janhunen, Juha. 2012. *Mongolian*. Amsterdam/Philadelphia: John Benjamins.
- Key, Gregory. 2013. The morphosyntax of the Turkish causative construction. Doctoral dissertation, University of Arizona.
- Kullman, Rita and Dandi-Ydam, Tserenpil. 2015. *Mongolian Grammar*, 5th edition, Switzerland: Kullnom Verlag.
- Kural, Murat. 1996. Verb incorporation and elementary predicates. Doctoral dissertation, The University of California.
- Miyagawa, Shigeru. 1999. Causatives. In Natsuko Tsujimura (eds.), *The Handbook of Japanese*

- Linguistics*, 236–268. Malden, Massachusetts, and Oxford, UK: Blackwell Publishers.
- Nie, Yining. 2020b. Morphological causatives are Voice over Voice. *Word Structure* 13:102–126.
- Nie, Yining. 2025. Turkish causatives are recursive: A response to Key 2013. *Linguistic Inquiry* 56:401–414.
- Pylkkänen, Liina. 2008. *Introducing Arguments*. Cambridge, Ma.: MIT Press.
- Shibatani, Masayoshi. 1976. The grammar of causative constructions: A conspectus. In Masayoshi Shibatani (eds.), *Syntax and Semantics: The Grammar of Causative Constructions*, 1–39. London: Academic Press.
- Sigurðsson, Einar Freyr, and Jim Wood. 2021. On the implicit argument of Icelandic indirect causatives. *Linguistic Inquiry* 52:579–626.